BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Performance

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is FTP used for?

2. How does phishing trick people?

A) Tries every possible password
B) Floods a server with traffic
C) Repairs damaged files
D) Pretends to be trusted

3. "Bluetooth is normally the best choice for a high-capacity link between distant school sites." Correct this. Explain why.

4. A network shows 2001:db8::7. "IPv6 uses four decimal numbers separated by dots." Correct this. How is this address written?

Today's objectives

Explain the core ideas of network performance using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Bandwidth
The available data transfer capacity of a connection over time.

Latency
Delay before data begins or a response is received.

Wired and wireless
Wired links tend to be less affected by wireless interference; wireless provides mobility.

Performance factors
Traffic, distance, medium, devices, interference and service contention affect observed speed.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Bandwidth	The available data transfer capacity of a connection over time.
Latency	Delay before data begins or a response is received.
Wired and wireless	Wired links tend to be less affected by wireless interference; wireless provides mobility.
Performance factors	Traffic, distance, medium, devices, interference and service contention affect observed speed.

I Do - One worked example

A 24 Mb file travels at a constant 8 Mb/s. Calculate ideal transfer time.

1. Identify units: both are megabits.
2. Use time = file size / transfer rate.
3. Compute 24 / 8 = 3 seconds.
4. Explain this ideal figure excludes latency/overhead.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A 60 Mb file travels at 12 Mb/s. Calculate ideal time.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the calculation of transfer time in seconds.

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb * speedMbps
print(seconds)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. Explain the difference between bandwidth and latency. [2 marks]

Q3. Calculate time for a 40 Mb file over 10 Mb/s ideal link. [2 marks]

Q4. Give two factors reducing Wi-Fi performance. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Bandwidth and latency are exactly the same measurement. ☐ T ☐ F
2. The available data transfer capacity of a connection over time. ☐ T ☐ F
3. Delay before data begins or a response is received. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Transferring files between networked computers. ☐

Q2 D - Pretends to be trusted ☐

Q3 Bluetooth suits short-range links, such as keyboards or headphones. Check range, data rate and power needs. ☐

Q4 IPv6 uses hexadecimal groups separated by colons. Groups of zeros can be shortened. Routers use IP addresses between networks; switches use MAC addresses within a LAN. ☐

Code First - Model

```
sizeMb = 90
speedMbps = 30
seconds = sizeMb / speedMbps
print(seconds)
```

Output: 3

We Do - Check the model

1. 5 seconds. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Bandwidth is data capacity per unit time; latency is delay before response/transfer. ☐

Q3 40 / 10 = 4 seconds. ☐

Q4 Interference, weak signal/distance, number of users or obstacles. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: